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| 主 题: | ICASSP2025 : Paper 434              |                     |
| 发件人: | "Microsoft CMT" <email@msr-cmt.org> | 2024-11-11 22:33:45 |
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Shihong Duan has uploaded review for Separable Large-Kernel Graph Convolution with Hierarchical Aggregation Transformer for Skeleton-based Action Recognition

-- Review Summary --

1. How confident are you in your evaluation of this paper?
    2. Confident
  2. Importance/Relevance
    3. Of sufficient interest
  3. Classify the type of paper
    - Neutral
  4. Originality/Novelty
    2. Minor originality; provides barely sufficient new insight or understanding
  5. Justification of Originality/Novelty Score (required)
 

The originality lies in the combinational innovation of the method.
  6. Theoretical Development
    3. Probably correct; provides limited new insights or understanding
  7. Experimental Validation
    2. Lacking in some respect
  8. Justification of Experimental Validation Score (required if score is 1 or 2).
 

The ablation experiments are insufficient, and the selection of experimental hyperparameters is unclear.
  9. Clarity of Presentation
    3. Clear enough
  10. Reference to Prior Work
    3. References adequate
  11. Overall evaluation of this paper
    2. Marginal reject
  12. Justification of Overall evaluation of this paper (required)
    - 1.The abstract mentions the proposed SLK-GCN, a separable large-kernel graph convolutional network, but it would be beneficial to highlight its ability to effectively model long-range spatial configurations. On which benchmarks are the experiments conducted? Please be as specific as possible. Additionally, the terms "effectiveness and superiority" require concrete performance metrics to make the claims clearer. The same issue appears in the description of the main contributions. It is recommended to include the theoretical principles behind the method to demonstrate its value and add specific performance metrics to validate its effectiveness.
    - 2.In Chapter 1, add a brief discussion of application scenarios to clarify the problem and model context.
    - 3.The title of Figure 2 is "Hierarchical Aggregation Transformer," but the figure does not clearly illustrate the hierarchical aggregation process. The hierarchical structure is not evident, and it is unclear how the Transformer module aggregates information. The input-output relationships are not clearly defined, and the role of the MLP should also be explicitly stated.
    - 4.The description of the Local-and-Global Temporal Module (LAG-TT) is unclear. What is the purpose of "concatenating the k segments end-to-end"? How does this operation ensure the effective capture of global temporal information? Additionally, how is the value of k determined?
    - 5.The specific interactions and dependencies between the GCN stream and the Transformer stream are not thoroughly explained.
    - 6.The complementary nature of the multi-stream structure lacks quantitative analysis, and the ablation experiments could be more detailed. For the channel configuration in the model architecture, was any optimization performed? There is no explanation for the selection of these parameters.
  13. Award Quality (only for papers marked "Definite accept")
    - 0: Not award quality
  14. Confidential Comments to technical program committee (will not be seen by authors):
 

The method is not clear enough, the explanation of the experimental results is insufficient, and the basis for hyperparameter selection is unclear.
  15. All reviewers must accept and agree to follow IEEE Policies. By submitting your review, you acknowledge that you have read and agree to IEEE's Privacy Policy (<https://www.ieee.org/security-privacy.html>).
- Agreement accepted

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